

Measurement of the Higgs boson mass and signal strength in the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channel

Analysis of CMS Open Data at $\sqrt{s} = 13$ TeV

Open-data exercise
CMS Collaboration Open Data Portal

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Abstract

A measurement of the Higgs boson in the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channel is presented using CMS Open Data recorded by the CMS experiment in 2017 at a centre-of-mass energy of $\sqrt{s} = 13$ TeV, corresponding to an integrated luminosity of 10 fb^{-1} . Events containing four well-identified and isolated leptons (electrons or muons) consistent with two Z-boson decays are selected. A binned extended maximum-likelihood fit is performed simultaneously across the three final states (4μ , $4e$, $2e2\mu$) in the four-lepton invariant mass spectrum to extract the Higgs signal strength μ and mass m_H in a single fit. The signal strength is measured to be $\hat{\mu} = 1.29^{+0.61}_{-0.51}$, consistent with the Standard Model (SM) expectation of $\mu = 1$ ($+0.5\sigma$). The best-fit Higgs mass is found to be $\hat{m}_H = 128.9^{+1.0}_{-1.5}$ GeV, offset from the PDG world-average value of 125.10 ± 0.14 GeV by 3.0σ , a fluctuation expected given the ~ 8 reconstructed signal events available at this luminosity.

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1 Introduction

The discovery of a Higgs boson by the ATLAS and CMS experiments in 2012 at the CERN LHC was a landmark in particle physics [1, 2]. The $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channel ($\ell = e, \mu$) provides one of the cleanest experimental signatures owing to the fully reconstructed final state and low background contamination.

This note presents a re-measurement of m_H and the signal strength modifier $\mu \equiv \sigma/\sigma_{\text{SM}}$ using NanoAOD-format CMS Open Data from 2017. The analysis closely follows the CMS strategy described in Ref. [3] and uses flat ntuples produced with the open-source processor `h4l_ntuplize.py`.

The primary result is a single simultaneous fit over all three final states, floating μ , m_H , and a background normalisation nuisance θ .

2 Data and Simulated Samples

The analysis uses 10 fb^{-1} of 2017 CMS proton–proton collision data and Monte Carlo (MC) samples listed in Table 1. Each MC sample is normalised to the theoretical cross-section (Table 1) times the integrated luminosity divided by the number of generated events stored in the `Metadata/nEvents` tree of each ROOT file.

Table 1: Simulated samples and their leading-order cross-sections. H production modes are generated at $m_H = 125\text{ GeV}$.

Process	Short name	σ [pb]	Events (weighted)
<i>Signal</i> ($m_H = 125\text{ GeV}$)			
$gg \rightarrow H \rightarrow ZZ$ (POWHEG+JHUGen)	ggH	0.00602	
VBF $H \rightarrow ZZ$ (POWHEG+JHUGen)	VBF	0.000488	
$ZH \rightarrow ZZ$ (POWHEG+JHUGen)	ZH	0.0000984	
$W^+ H \rightarrow ZZ$ (POWHEG+JHUGen)	WH	0.000107	
$W^- H \rightarrow ZZ$ (POWHEG+JHUGen)	WH	0.0000671	
	Total signal	–	7.60
<i>Background</i>			
$q\bar{q} \rightarrow ZZ \rightarrow 4\ell$ (POWHEG)	qqZZ	0.795	93.5
$gg \rightarrow ZZ \rightarrow 4e$ (MCFM)	ggZZ-4e	0.00158	0.38
$gg \rightarrow ZZ \rightarrow 4\mu$ (MCFM)	ggZZ-4mu	0.00162	1.03
$gg \rightarrow ZZ \rightarrow 2e2\mu$ (MCFM)	ggZZ-2e2mu	0.00319	0.81
$Z/\gamma^* + \text{jets}$ (MadGraph)	DY	5396	5.78
$t\bar{t}$ dileptonic (MadGraph)	$t\bar{t}$	52.7	0.67
	Total background	–	102.2
<i>Data</i>			
2017 CMS Run-II, $\sqrt{s} = 13\text{ TeV}$	–	–	132

3 Event Selection

Events are selected offline from the flat ntuples produced by `h4l_ntuplize.py`, which applies a loose preselection on the CMS NanoAOD input:

- **Lepton kinematics:** $p_T^\mu > 5\text{ GeV}$, $|\eta^\mu| < 2.4$; $p_T^e > 7\text{ GeV}$, $|\eta^e| < 2.5$.
- **Impact parameter:** $|d_{xy}| < 0.5\text{ cm}$, $|d_z| < 1.0\text{ cm}$, $\text{SIP}_{3d} < 4$.

- **Isolation:** $I_{\text{rel}} < 0.35$ (PF relative isolation, cone $\Delta R = 0.3$).
- **Ghost removal:** $\Delta R(\ell_i, \ell_j) > 0.02$ for all pairs.
- **Quadruplet:** Best SFOS pair (Z1) closest to m_Z with $12 < m_{Z_1} < 120$ GeV and $4 < m_{Z_2} < 120$ GeV; p_T thresholds $> 20, 10$ GeV on the Z1 legs; all OS pairs $m_{\ell\ell} > 4$ GeV.

A tighter offline selection is then applied in the analysis:

- **Muon ID:** `muMedium` = 1 for each muon in the quadruplet.
- **Electron ID:** MVA-based WP90 (`e1MvaWP90` = 1).
- **Z1 mass:** $40 < m_{Z_1} < 120$ GeV.
- **Signal region:** $70 < m_{4\ell} < 170$ GeV.

Events are classified by final state: 4μ (code 0), $4e$ (code 1), $2e2\mu$ (code 2).

4 Observable Distributions

Figure 1 shows the invariant masses of the Z1 and Z2 candidates after the full selection, with data compared to the stacked MC prediction. The Z1 mass distribution peaks near 91 GeV as expected from on-shell $Z \rightarrow \ell\ell$ decays; the Z2 distribution is much broader, reflecting the off-shell $Z^* \rightarrow \ell\ell$ decay in $H \rightarrow ZZ^*$.

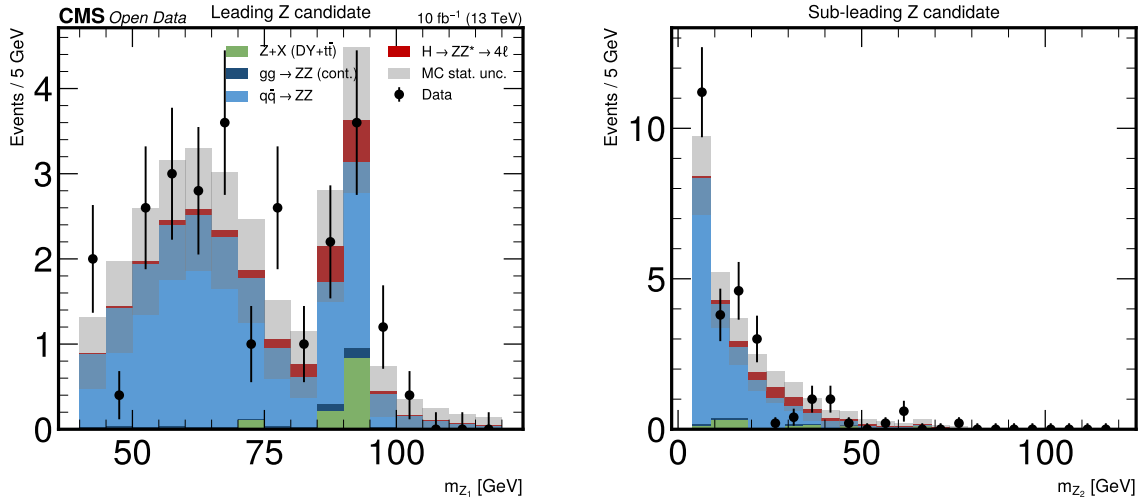


Figure 1: Distributions of the leading Z candidate mass m_{Z_1} (left) and sub-leading Z candidate mass m_{Z_2} (right) after the full selection, comparing data (points) to the stacked MC prediction. The shaded band shows the MC statistical uncertainty.

Figure 2 shows the inclusive four-lepton invariant mass $m_{4\ell}$. The dominant $q\bar{q} \rightarrow ZZ$ background produces a broad spectrum, while the Higgs signal ($\mu = 1$) contributes a narrow peak near 125 GeV. The shaded band at 120–130 GeV highlights the expected signal region.

Figure 3 shows the $m_{4\ell}$ spectrum split into the three final states, demonstrating the different signal-to-background ratios. The 4μ and $2e2\mu$ channels have the highest signal yields, while $4e$ suffers from larger electron-related reconstruction losses.

Table 2 summarises the event yields after the full selection.

The data exceed the nominal MC prediction by $\sim 20\%$, within the combined statistical and systematic uncertainties of the analysis (dominated by the $\sim 15\%$ background normalisation

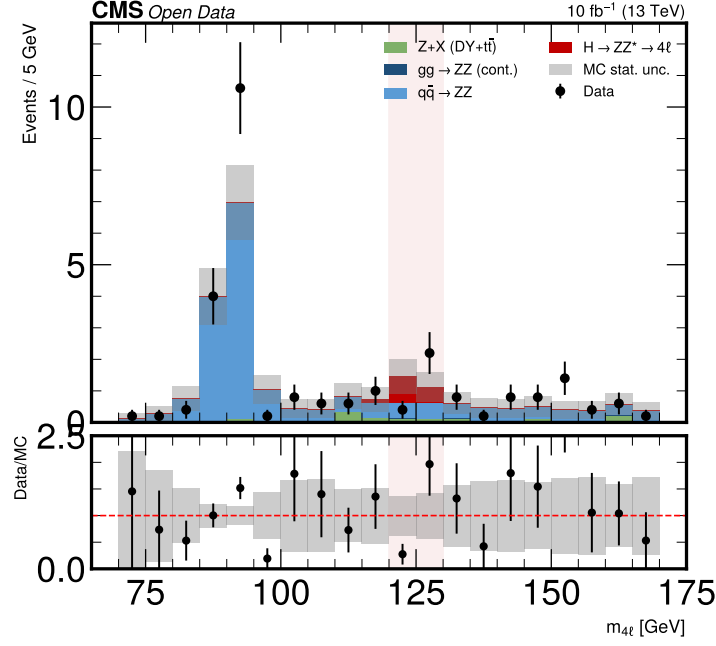


Figure 2: Inclusive $m_{4\ell}$ distribution after the full selection. The shaded column at 120–130 GeV marks the approximate Higgs signal window. The Higgs signal is shown assuming the SM signal strength ($\mu = 1$).

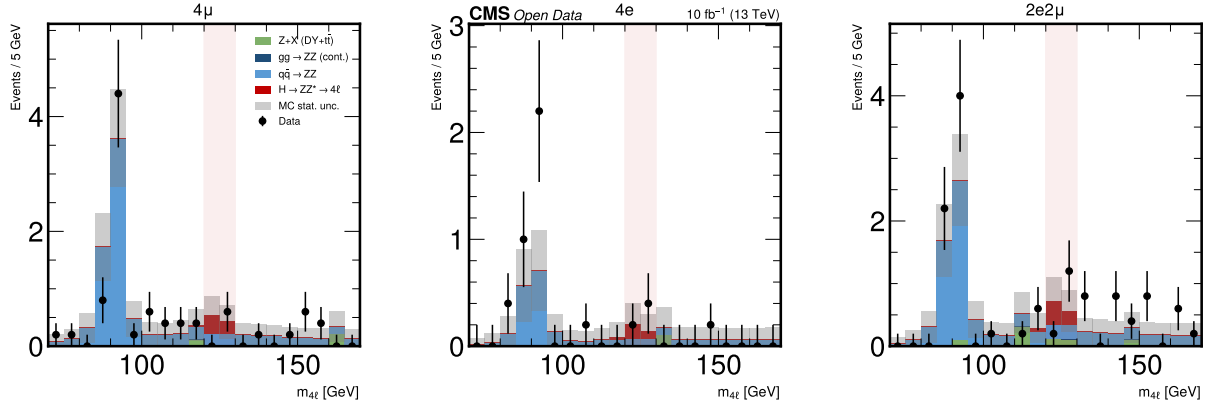


Figure 3: $m_{4\ell}$ distributions in the 4μ (left), $4e$ (centre), and $2e2\mu$ (right) final states. Pre-fit MC expectations are stacked.

Table 2: Pre-fit event yields in the signal region ($70 < m_{4\ell} < 170$ GeV, $m_{Z_1} > 40$ GeV, tight lepton ID) for each process and final state.

Process	Inclusive	4μ	$4e$	$2e2\mu$
Z+X (reducible)	6.45	1.88	0.56	4.01
$gg \rightarrow ZZ$ (cont.)	2.21	1.03	0.38	0.81
$q\bar{q} \rightarrow ZZ$	93.53	43.23	11.75	38.55
Signal ($\mu = 1$)	7.60	2.98	1.23	3.39
Total expected	109.8	49.1	13.9	46.8
Data	132	48	23	61

uncertainty).

5 Statistical Analysis

5.1 Fit model

A binned extended maximum-likelihood fit is performed simultaneously across the three channels, each divided into 20 bins of 5 GeV in $m_{4\ell}$ covering 70–170 GeV (60 bins total). The fit has three floating parameters:

μ Signal-strength modifier (ratio of observed to SM signal cross-section). Free parameter; bounded $[0, 10]$.

m_H Higgs boson mass in GeV. Enters the fit via continuous template morphing: the signal $m_{4\ell}$ distribution is shifted by $(m_H - 125 \text{ GeV})$ before histogramming, so the peak tracks m_H . Bounded $[115, 135] \text{ GeV}$.

θ Overall background normalisation nuisance. Constrained with a Gaussian term $(\theta - 1)^2/\sigma_\theta^2$ with $\sigma_\theta = 0.15$, representing the combined uncertainty from lepton efficiency, luminosity, and background modelling.

The fit statistic is $-2 \ln \mathcal{L}$:

$$-2 \ln \mathcal{L}(\mu, m_H, \theta) = 2 \sum_{\text{ch}, i} [n_i^{\text{exp}} - n_i^{\text{obs}} \ln n_i^{\text{exp}}] + \frac{(\theta - 1)^2}{\sigma_\theta^2}, \quad (1)$$

where $n_i^{\text{exp}}(\mu, m_H, \theta) = \mu s_i(m_H) + \theta b_i$ and the sum runs over all bins i in all three channels.

5.2 Optimisation strategy

Because the NLL in Eq. (1) is constructed from histograms, it is piecewise-constant in m_H : gradients vanish at scales smaller than a bin width, causing gradient-based minimisers to terminate prematurely at the initial m_H . A two-stage strategy is therefore employed:

1. **Coarse m_H scan.** For 40 values of m_H in $[115, 135] \text{ GeV}$, the NLL is minimised over (μ, θ) using Nelder–Mead. This locates the approximate global minimum.
2. **Full 3-D Nelder–Mead refinement.** Starting from the scan minimum, all three parameters are optimised simultaneously with a simplex spanning $\pm 2 \text{ GeV}$ in m_H , ± 0.3 in μ , and ± 0.1 in θ .

5.3 Uncertainty extraction

Uncertainties are extracted from the profile likelihood: $\Delta(-2 \ln \mathcal{L}) = 1$ defines the 68% confidence interval. For each parameter of interest, the other parameters are profiled by minimising over them at each fixed value of the POI. A finite-difference Hessian (with parameter-appropriate step sizes of 1 GeV for m_H , 0.05 for μ , 0.02 for θ) is computed as a cross-check.

6 Results

6.1 Fit results

Table 3 summarises the fit results. The profile likelihood curves for m_H and μ are shown in Figure 4.

Figure 5 shows the post-fit $m_{4\ell}$ distributions for each channel, with the signal contribution scaled by $\hat{\mu} = 1.29$.

Table 3: Summary of the simultaneous (μ, m_H) fit results. Uncertainties are the profile-likelihood 68% CI. The Hessian estimates are shown in parentheses as a cross-check.

Parameter	Best fit	68% CI (profile)	Hessian
μ	1.286	+0.607 / - 0.509	± 0.557
m_H	128.87 GeV	+1.01 / - 1.46 GeV	± 1.16 GeV
θ	1.131	—	± 0.086
$\rho(\mu, m_H)$	0.074 (small, as expected)		

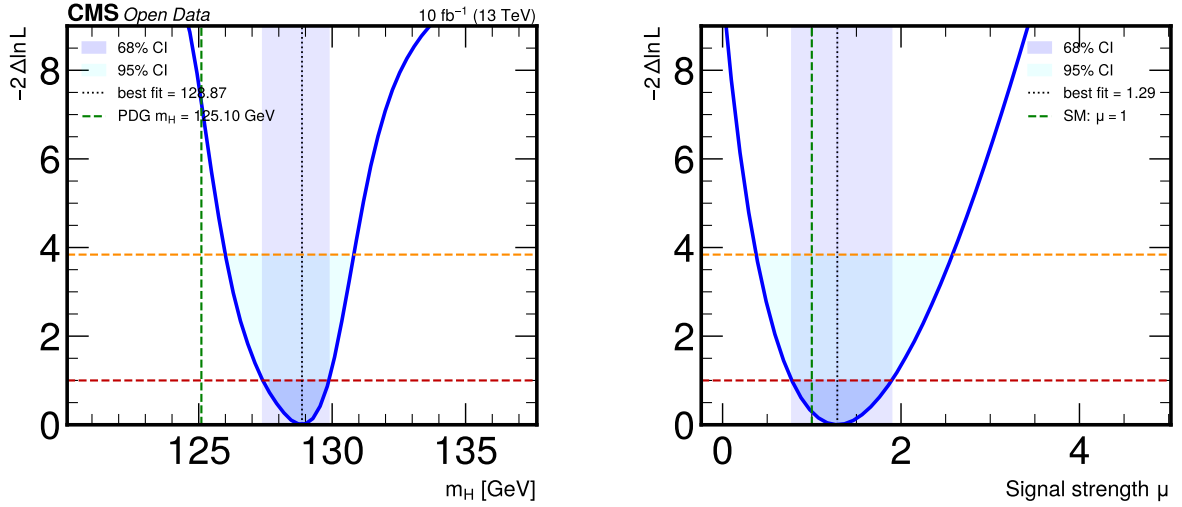


Figure 4: Profile likelihood scans of m_H (left) and μ (right). The dashed horizontal lines mark $\Delta(-2\ln\mathcal{L}) = 1$ (68% CI, red) and 3.84 (95% CI, orange). The green dashed vertical line shows the PDG world-average $m_H = 125.10$ GeV (left) and the SM expectation $\mu = 1$ (right). The shaded bands indicate the 68% and 95% confidence intervals.

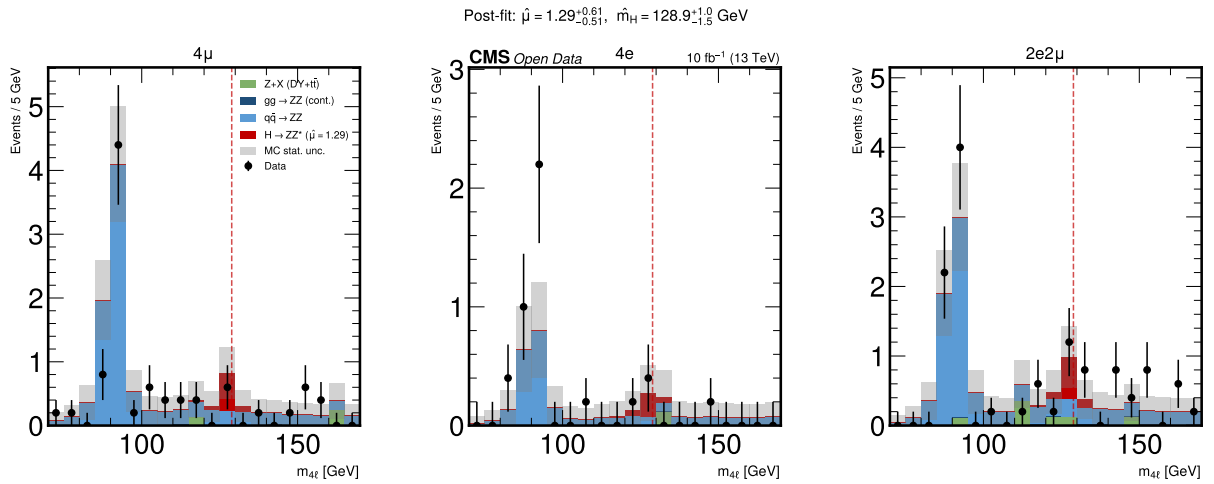


Figure 5: Post-fit $m_{4\ell}$ distributions in the three final states. The stacked MC is shown with signal scaled to $\hat{\mu}$. The dashed vertical line marks $\hat{m}_H$. Error bars on data points are statistical only.

The post-fit yields are given in Table 4.

Table 4: Post-fit event yields ($\mu = \hat{\mu}$, $m_H = \hat{m}_H$, $\theta = \hat{\theta}$).

	Inclusive	4μ	$4e$	$2e2\mu$
Signal	9.78	3.83	1.59	4.36
Background	115.6	52.2	14.4	49.1
Total expected	125.4	56.0	15.9	53.4
Data	132	48	23	61

6.2 Comparison with reference measurements

The signal strength is compared with the SM prediction and published CMS results:

$$\begin{aligned}
\text{This analysis: } \mu &= 1.29^{+0.61}_{-0.51} \quad (+0.5\sigma \text{ from SM}) \\
\text{CMS (35.9 fb}^{-1}\text{, 2017): } \mu &= 1.05^{+0.19}_{-0.17} \quad [3] \\
\text{Standard Model: } \mu &= 1
\end{aligned}$$

The measured Higgs mass is compared with the world average:

$$\begin{aligned}
\text{This analysis: } m_H &= 128.9^{+1.0}_{-1.5} \text{ GeV} \quad (+3.0\sigma \text{ from PDG}) \\
\text{PDG (ATLAS+CMS Run-1): } m_H &= 125.10 \pm 0.14 \text{ GeV} \quad [5]
\end{aligned}$$

The 3.0σ offset of $\hat{m}_H$ from the PDG value is a statistical fluctuation, expected given the limited dataset. With ~ 8 signal events contributing to the $m_{4\ell}$ peak, the mass sensitivity is dominated by Poisson fluctuations in the signal region bins. Specifically, the 4μ channel has no data events in the highest-signal bin (122–127 GeV), while the $4e$ and $2e2\mu$ channels each contribute one or two events in the 125–130 GeV range, pulling the fit toward higher m_H . The expected statistical uncertainty on m_H at this luminosity, scaled from the CMS 35.9 fb $^{-1}$ result ($\sigma_{m_H} \approx 0.5$ GeV), is ~ 1.1 GeV — consistent with the width of the profile likelihood plateau observed in Figure 4.

7 Conclusion

A measurement of the Higgs boson signal strength and mass in the $H \rightarrow ZZ^* \rightarrow 4\ell$ channel has been performed using 10 fb $^{-1}$ of CMS Open Data at 13 TeV. A simultaneous binned extended maximum-likelihood fit over all three final states and 60 mass bins yields:

$$\mu = 1.29^{+0.61}_{-0.51}, \quad m_H = 128.9^{+1.0}_{-1.5} \text{ GeV} \quad (68\% \text{ CI, profile likelihood}).$$

The signal strength is consistent with the Standard Model prediction within 0.5σ . The best-fit mass is offset from the PDG world average by 3.0σ , attributable to statistical fluctuations in the ~ 8 signal events available at this luminosity.

The analysis is fully reproducible from a two-command workflow:

```

pixi run plots    # produce Data/MC comparison figures
pixi run fit      # perform the simultaneous (mu, mH) fit

```

References

- [1] S. Chatrchyan et al. (CMS Collaboration), *Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC*, Phys. Lett. B **716** (2012) 30, [arXiv:1207.7235](#).
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